ATMT Assignment 4

Topic: Exam Preparation

Deadline: ~~Tuesday, November 18, 2025, 14:00~~
Thursday, November 20, 2025, 18:00

Important submission info:

- **Submission format: pdf**
- Please name your submission files like this:
  olatusername1_(olatusername2_)atmt_assignmentxx.pdf,
  e.g. `mmuster_jsmith_atmt_assignment03.pdf`
- For this assignment, you are asked to submit a report as a PDF file. Please do not hand in any other files.
- You need to submit the assignment on OLAT using the tab for assignment 4. Please hand it in on time. The submission deadline is given above. After this time, the tab will be closed.
- You are encouraged to work in groups of two. Please make sure every file clearly states **both** names. For pairs, only one person should submit the assignment.

Assignment Description

This assignment is intended to help you prepare for the exam. There are 8 topical sections with questions that you need to answer. Some of these questions are taken from previous exams.[1] You should be able to answer these questions with what you learned in the lectures so far.

You must answer at least 3 sections correctly to pass this assignment with a score of 0.5. You must answer at least 6 sections correctly to pass with a score of 1. If you do not answer all questions of a section correctly, the section will **not** count as solved.

Generally, you should aim to write 4 to 5 sentences for every question unless stated otherwise. There will not be any official solutions for this assignment but you will get individual feedback on the questions you solved.

UPD: due to Kirill's trip to China and therefore change of the lecture plan, **section 7 is yet to be covered**. However, as stated above, answering 6 out of 8 sections correctly already gives you the full score, so you should not suffer for that. Just do not be afraid of the topic that was not yet covered.

1 True or False

For this question, say whether the following statements are true or false, and give a short justification for your answer.

a) The fact that a sentence can have multiple good translations makes training of neural machine translation systems easier.

b) A neural machine translation model cannot learn to disambiguate homographs[2]

c) Length normalization will increase the average length of translations.

2 Language Representations

a) How are words represented internally in a word-level neural language or translation model? Discuss both input and output words (no equations or diagrams are required).

b) What are the consequences (in terms of computational complexity) of increasing the network vocabulary size in neural machine translation?

[1]Note that the actual exam will have similar questions, but will be shorter.
[2]words with the same spelling, but different meanings, such as *fine* (thin; penalty)

3 Sequence-to-Sequence Models

a) Explain why an encoder-decoder model is essential for the task of translation compared to language modelling. Use a concrete example and an illustration to support your answer.

b) Describe another NLP task where using a sequence-to-sequence model would **not** be appropriate. Why is this the case and how else would you model the task? What would be the inputs and outputs to your model?

4 Open Vocabulary Translation

a) What problem does subword-level NMT aim to solve and how?

b) Why might you want to use a smaller subword vocabulary size in a low-resource scenario?

c) What might be some of the problems related to sharing a single subword model across 100 languages in recently proposed massively multilingual neural machine translation models?

5 NMT Architectures

a) Complex recurrent units like the LSTM and GRU rely on *gates* to control the flow of information. What output range are gates typically intended to have, and what activation function(s) can be used to achieve this?

b) Name two "building blocks" of the Transformer and explain what each block does in 2 to 3 sentences.

c) Self-attention networks such as the Transformer use masking in the decoder to ignore any words or hidden states after the current time step. Peter experiments with this, and notices that his cross-entropy at training time improves when he removes masking. Why is masking used, and what are the consequences of removing it?

6 Learning from Monolingual or Multilingual Data

a) Briefly summarize the idea behind trying to leverage monolingual data in NMT and explain the differences between the following two techniques that aim to do this:

- Combining MT with Autoencoding

- Backtranslation

b) Give two motivations for multilingual NMT and provide examples to support your claims.

c) Describe how multi-source[3], multi-way[1] and massively multilingual models[2] differ and how do they aim to improve translation?

7 Advanced Training Objectives (Yet to be covered!)

a) Explain the following concepts in 2 to 3 sentences and describe how they are linked:

- Exposure bias
- Teacher forcing
- Minimum Risk Training

b) Explain the difference between label smoothing and word-level knowledge distillation. Use an example to illustrate your answer.

8 Linguistic Phenomena

a) You trained a sentence-based neural machine translation system from English to French. When looking through the translations you find that the model does not produce verb forms that consistently agree with the gender of the speaker. Describe a strategy for how you could retrain your system such that it allows you to control better which form is produced in the translation? What changes do you need to make to your system? What is a disadvantage of your chosen strategy?

b) How would you evaluate your new system? In which cases would you recommend an automatic strategy and in which cases a would you recommend human evaluation?

c) Do you think a system with more context than one sentence could handle this agreement better than a sentence-level system? What kind of context would you use in this case to train such a system?

References

[1] Orhan Firat, Kyunghyun Cho, and Yoshua Bengio. Multi-way, multilingual neural machine translation with a shared attention mechanism. In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 866–875, San Diego, California, June 2016. Association for Computational Linguistics.

[2] Melvin Johnson, Mike Schuster, Quoc V. Le, Maxim Krikun, Yonghui Wu, Zhifeng Chen, Nikhil Thorat, Fernanda Viégas, Martin Wattenberg, Greg Corrado, Macduff Hughes, and Jeffrey Dean. Google's multilingual neural machine translation system: Enabling zero-shot translation. *Transactions of the Association for Computational Linguistics*, 5:339–351, 2017.

[3] Barret Zoph and Kevin Knight. Multi-source neural translation. In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 30–34, San Diego, California, June 2016. Association for Computational Linguistics.

Please let us know in the OLAT forum if you have problems or something is unclear.
Good luck!